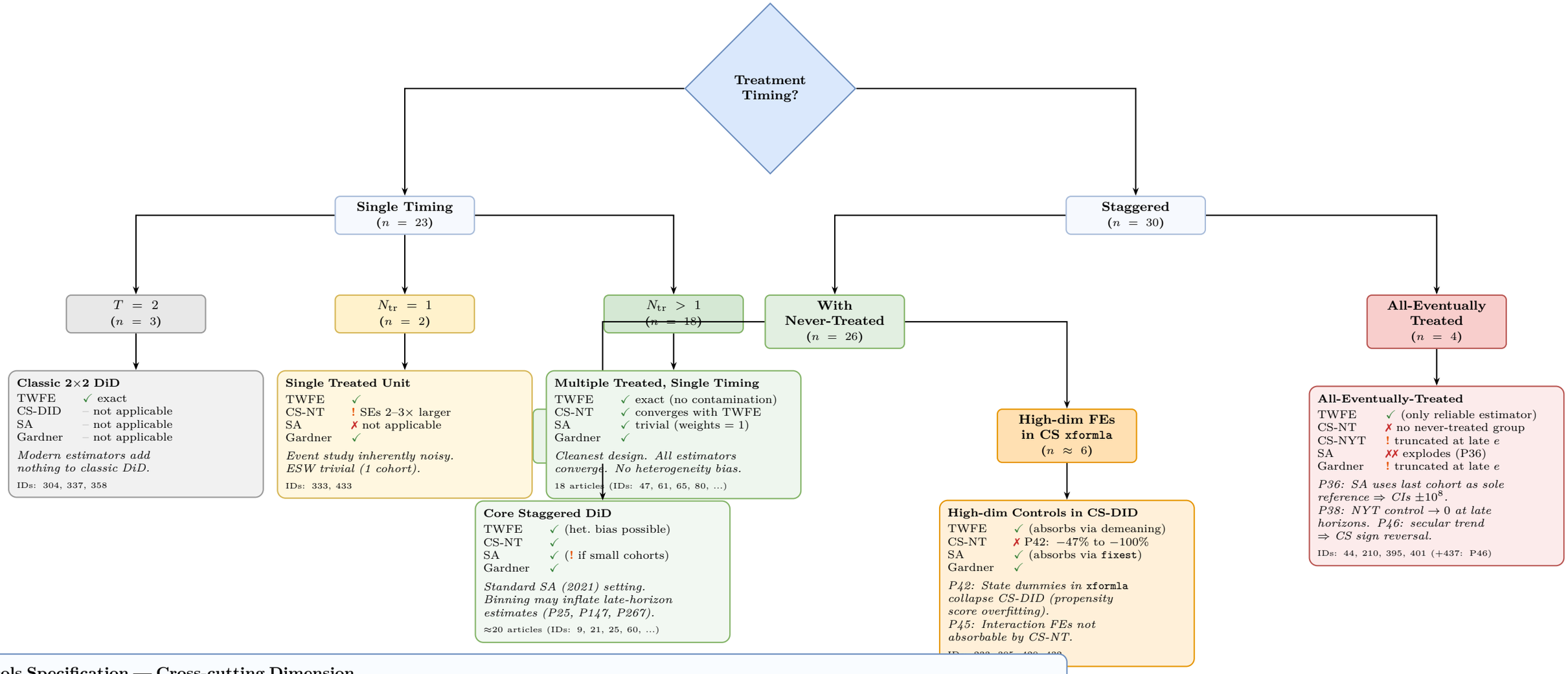

Controls Specification — Cross-cutting Dimension

Control Type	TWFE/SA/Gardner	CS-DID	Empirical Finding
No controls	✓	✓	All estimators unaffected
Simple time-varying	✓	✓	CS-DID OK if low-dimensional (`xformla = ~ x1 + x2`)
State/region dummies in `xformla`	✓	✗ P42	CS-DID collapses −47% to −100%; propensity score overfitting (5/8 articles)
Interaction FEs (state×year)	✓	✗ P45	CS-NT pre-trends diverge (+45 vs ≈ 0); conditional PT assumption exposed
Baseline × time trends	✓	! P37	Not time-varying covariates; include as controls, not in `xformla`

✓ = works ! = limited/caution ✗ = fails or unreliable

P = Pattern number (see text). n = articles in sample.
Based on 53 replicated health economics articles.